Differential Equations and Slope Fields

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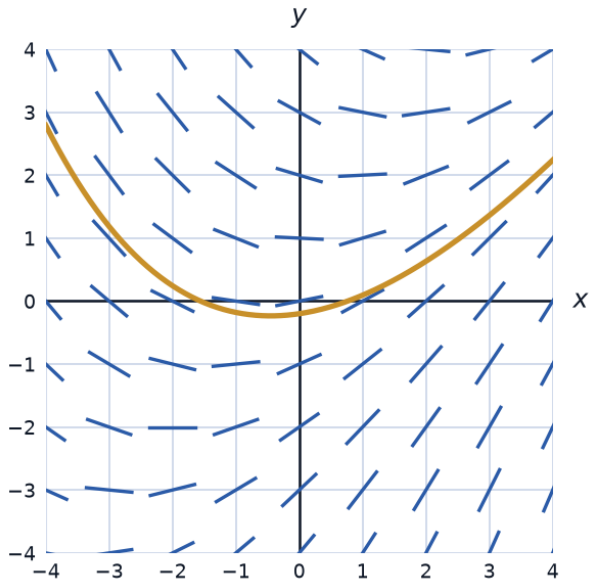
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Verifying solutions to differential equations

Verify that $y = Ce^{3x}$ is a solution to the differential equation $\frac{dy}{dx} = 3y$ for any constant C.

2 [Foundational] • Target: Verifying solutions to differential equations

Verify that $y = x^2 + C$ is a general solution to $\frac{dy}{dx} = 2x$, and find the particular solution satisfying $y(1) = 5$.



3 [Foundational] • Target: Solving separable differential equations

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.

4 [Foundational] • Target: Solving separable differential equations

Solve $\frac{dy}{dx} = 2xy$ given $y(0) = 1$.

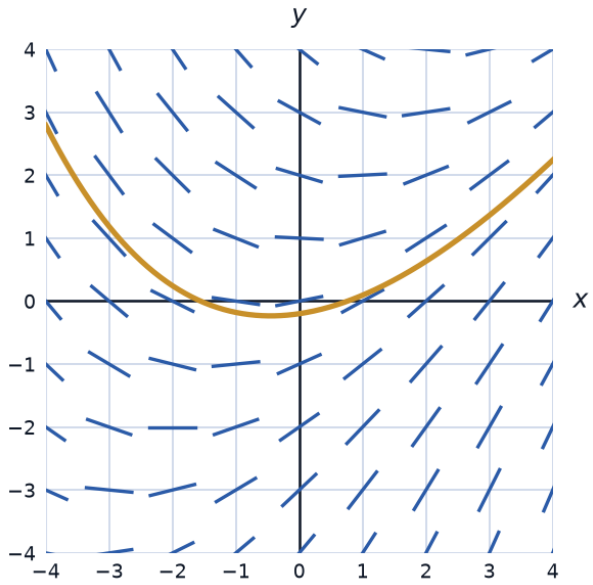
5 [Foundational] • Target: Solving separable differential equations

Solve $\frac{dy}{dx} = \frac{\cos x}{y^2}$ given $y(0) = 2$.

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6 [Foundational] • Target: Slope fields

For the differential equation $\frac{dy}{dx} = x - y$, find the slope at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$, as would be drawn on a slope field.



7 [Foundational] • Target: Slope fields

Explain how a slope field for $\frac{dy}{dx} = ky$ (exponential growth/decay) would look different for $k > 0$ versus $k < 0$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Exponential growth and decay models

A bacteria culture grows according to $\frac{dP}{dt} = 0.2P$, with $P(0) = 500$. Find $P(t)$ and the population after 10 hours.

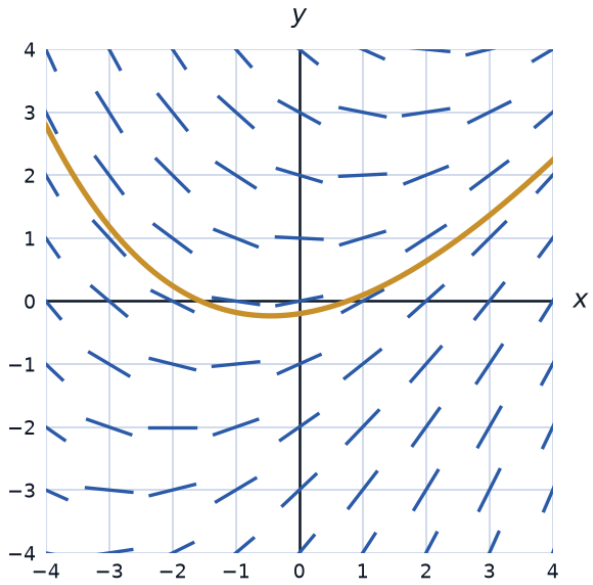
9 [Intermediate] • Target: Exponential growth and decay models

A radioactive substance decays according to $\frac{dA}{dt} = -0.05A$. If $A(0) = 100$ grams, find how long it takes to decay to 50 grams (the half-life).

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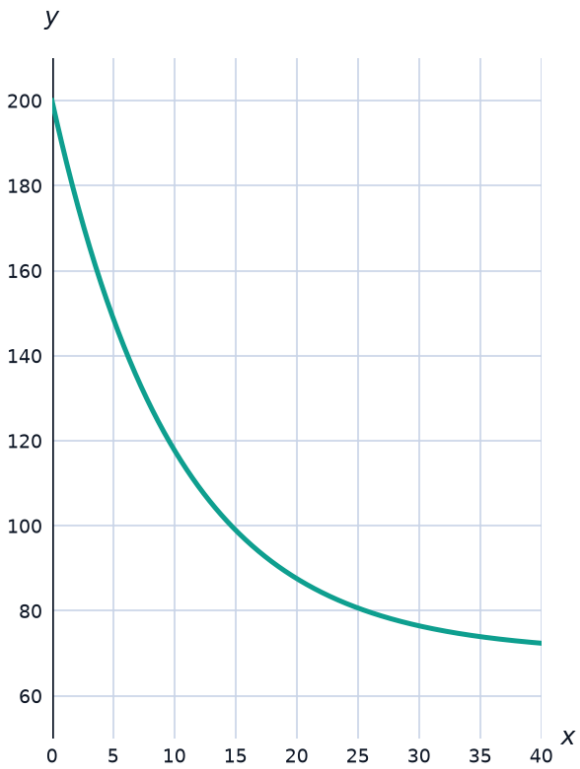
10 [Intermediate] • Target: Matching slope fields to differential equations

A slope field shows horizontal segments (slope 0) along the entire x-axis and along the entire y-axis, with slopes elsewhere resembling the product xy. Identify the differential equation this slope field represents.



11 [Intermediate] • Target: Newton's Law of Cooling

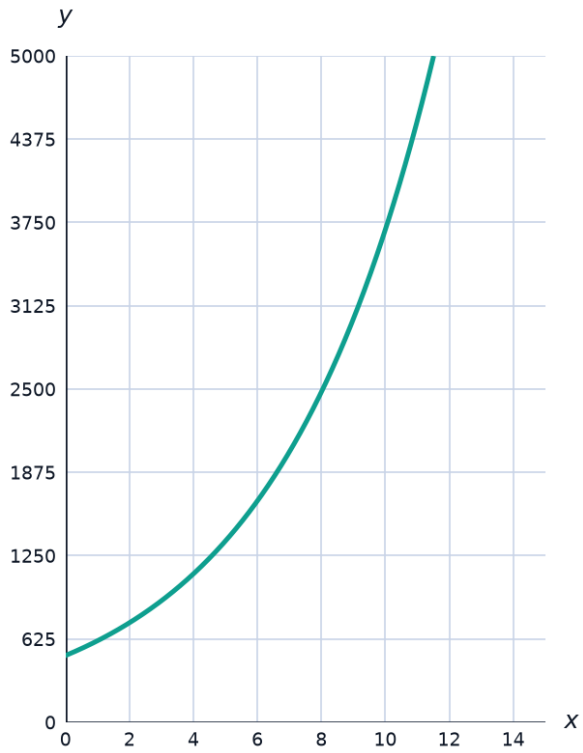
An object cools according to $\frac{dT}{dt} = -k(T - 70)$, where $70°\text{F}$ is room temperature. If $T(0) = 200°\text{F}$ and $k = 0.1$, find $T(t)$.



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12 **[Intermediate]** • **Target: Visualizing an exponential growth solution**

The graph shows the bacteria population $P(t) = 500e^{0.2t}$ from the earlier problem. Use the graph to estimate how long it takes for the population to double from its initial value of 500, then verify algebraically using $\ln 2$.



13 **[Intermediate]** • **Target: A logistic-style differential equation**

A population satisfies $\frac{dP}{dt} = 0.1P\left(1 - \frac{P}{200}\right)$. Find the equilibrium solutions (where $\frac{dP}{dt} = 0$), and explain which one is stable for a population starting near $P = 10$.

14 **[Intermediate]** • **Target: Additional separable equations**

Solve $\frac{dy}{dx} = \frac{y}{x}$ given $y(1) = 6$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

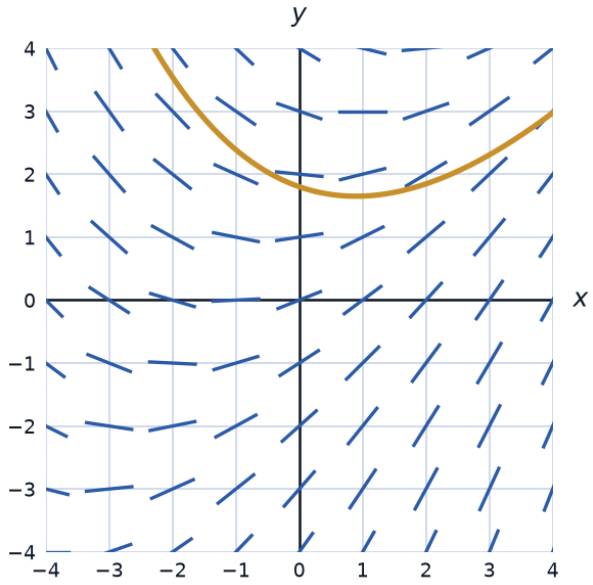
15 **[Challenging]** • **Target: Additional separable equations** • **Cross-Lesson: 8.01 Area Between Curves**

Solve $\frac{dy}{dx} = \frac{1}{y}$ given $y(0) = 4$.

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16 [Challenging] • Target: A mixing problem modeled by a differential equation • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

A tank contains 50 L of pure water. Brine with concentration 3 kg/L flows in at 2 L/min, and the well-mixed solution drains at the same rate. If $y(t)$ is the amount of salt (kg) at time t, the differential equation is $\frac{dy}{dt} = 6 - \frac{2y}{50} = 6 - \frac{y}{25}$. Find the equilibrium value of y, and explain what it represents physically.



17 [Challenging] • Target: Exponential growth and decay models • Cross-Lesson: 1.01 Limits Graphically and Numerically

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 20$. Find $y(15)$ (the positive root).

18 [Challenging] • Target: Matching slope fields to differential equations • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 12$. Find $y(-5)$ (the positive root).

19 [Challenging] • Target: Newton's Law of Cooling • Cross-Lesson: 2.01 The Definition of the Derivative

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 4$. Find $y(3)$ (the positive root).

20 [Challenging] • Target: Visualizing an exponential growth solution • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 24$. Find $y(-10)$ (the positive root).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Verifying solutions to differential equations

For the differential equation $\frac{dy}{dx} = -3x + 5y$, find the slope at the point $(-1, -4)$, as would be drawn on a slope field.

a. -19

b. -17

c. -15

d. 23

22 [Foundational] • Target: Solving separable differential equations

For the differential equation $\frac{dy}{dx} = -5x$, find the slope at the point $(0, -3)$, as would be drawn on a slope field.

a. -2

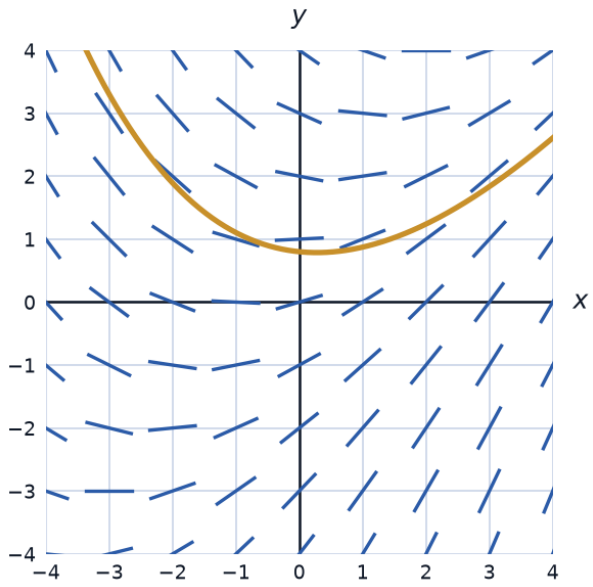
b. 1

c. 0

d. 2

23 [Foundational] • Target: Slope fields

For the differential equation $\frac{dy}{dx} = -x - 3y$, find the slope at the point $(3, 3)$, as would be drawn on a slope field.



a. 6

b. -14

c. -10

d. -12

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24 [Foundational] • Target: Exponential growth and decay models

For the differential equation $\frac{dy}{dx} = -3x + 5y$, find the slope at the point $(3, 1)$, as would be drawn on a slope field.

a. -4

b. -14

c. -6

d. -2

25 [Foundational] • Target: Matching slope fields to differential equations

For the differential equation $\frac{dy}{dx} = -5x + y$, find the slope at the point $(-2, -3)$, as would be drawn on a slope field.

a. 13

b. 7

c. 9

d. 5

26 [Foundational] • Target: Newton's Law of Cooling

For the differential equation $\frac{dy}{dx} = 0x + 2y$, find the slope at the point $(4, -4)$, as would be drawn on a slope field.

a. -10

b. 8

c. -8

d. -6

27 [Foundational] • Target: Visualizing an exponential growth solution

For the differential equation $\frac{dy}{dx} = -3x + 2y$, find the slope at the point $(-4, 4)$, as would be drawn on a slope field.

a. 20

b. 18

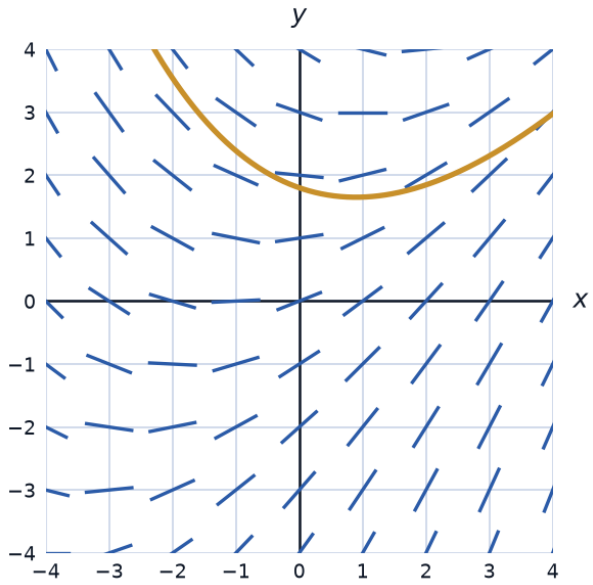
c. 4

d. 22

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28 [Foundational] • Target: A logistic-style differential equation

For the differential equation $\frac{dy}{dx} = -x + 2y$, find the slope at the point $(-4, -2)$, as would be drawn on a slope field.



a. 8

b. 0

c. -2

d. 2

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Additional separable equations

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 12$. Find $y(-9)$ (the positive root).

a. 12

b. 9

c. 14

d. 15

30 [Intermediate] • Target: A mixing problem modeled by a differential equation

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 40$. Find $y(-9)$ (the positive root).

a. 40

b. 39

c. 41

d. 9

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31 [Intermediate] • Target: Slope Fields & Solution Curves

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 12$. Find $y(9)$ (the positive root).

a. 9

b. 12

c. 15

d. 14

32 [Intermediate] • Target: Separation of Variables

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 7$. Find $y(-24)$ (the positive root).

a. 25

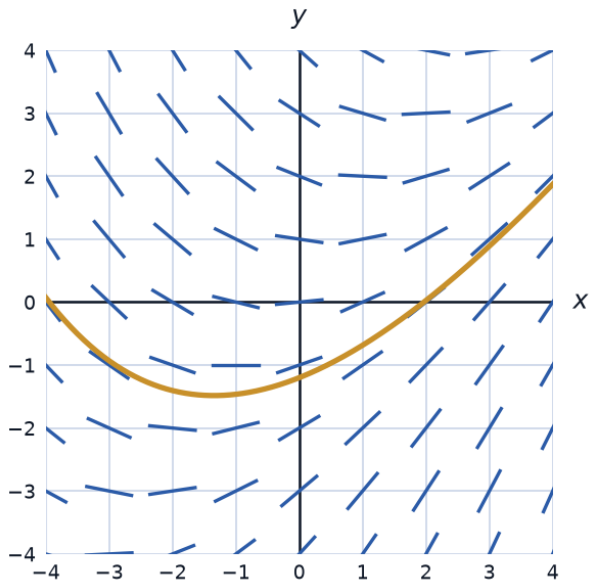
b. 28

c. 7

d. 24

33 [Intermediate] • Target: Exponential Growth & Decay Models

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 16$. Find $y(12)$ (the positive root).



a. 12

b. 19

c. 16

d. 20

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34 [Intermediate] • Target: Verifying solutions to differential equations

A quantity grows according to $\frac{dP}{dt} = \frac{3}{20}P$, with $P(0) = 425$. Find $P(12)$, rounded to the nearest whole number.

a. 2957

b. 2571

c. 2185

d. 2213

35 [Intermediate] • Target: Solving separable differential equations

A quantity grows according to $\frac{dP}{dt} = \frac{1}{4}P$, with $P(0) = 805$. Find $P(10)$, rounded to the nearest whole number.

a. 9807

b. 8336

c. 7638

d. 11278

36 [Intermediate] • Target: Slope fields

A substance decays according to $\frac{dA}{dt} = -\frac{1}{10}A$. If $A(0) = 126$ grams, find how long it takes to decay to 84 grams, to one decimal place.

a. 9.1

b. 2.8

c. 4.1

d. 5.3

37 [Intermediate] • Target: Exponential growth and decay models

A substance decays according to $\frac{dA}{dt} = -\frac{1}{20}A$. If $A(0) = 267$ grams, find how long it takes to decay to 89 grams, to one decimal place.

a. 22.0

b. 28.6

c. 27.0

d. 15.4

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Matching slope fields to differential equations • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 9 - \frac{y}{25}$. Find the equilibrium value of y.

a. 216

b. 34

c. 225

d. 250

39 [Challenging] • Target: Newton's Law of Cooling • Cross-Lesson: 2.01 The Definition of the Derivative

A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 10 - \frac{y}{30}$. Find the equilibrium value of y.

a. 300

b. 290

c. 330

d. 40

40 [Challenging] • Target: Visualizing an exponential growth solution • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 5 - \frac{y}{57}$. Find the equilibrium value of y.

a. 280

b. 62

c. 285

d. 342

41 [Challenging] • Target: Slope Fields & Solution Curves • Cross-Lesson: 3.01 The Chain Rule

Along which line in the slope field for $\frac{dy}{dx} = 2x + y$ are all slope segments horizontal?

a. $y = -2x$

b. $y = 2x$

c. $x = 0$

d. $y = 0$

42 [Challenging] • Target: Separation of Variables • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Solve the differential equation $\frac{dy}{dx} = 3x^2 y$ with initial condition $y(0) = 5$.

a. $y = 5e^{x^3}$

b. $y = e^{x^3} + 4$

c. $y = 5e^{3x^2}$

d. $y = x^3 + 5$

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43 [Challenging] • Target: Separation of Variables • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

What is the domain of the particular solution $y = \frac{1}{2-x}$ with $y(0) = \frac{1}{2}$?

a. $(-\infty, 2)$

b. $(2, \infty)$

c. $x \neq 2$

d. All real numbers

44 [Challenging] • Target: Exponential Growth & Decay Models • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

A population grows according to $\frac{dP}{dt} = 0.05P$. If initial population is $1,000$, what is the population at $t = 20$?

a. $1000e$

b. $1000e^2$

c. 2000

d. $1000(1.05)^{20}$

45 [Challenging] • Target: Newton's Law of Cooling • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Newton's Law of Cooling states that $\frac{dT}{dt} = -k(T - T_m)$. What does T_m represent?

a. The ambient temperature of the surrounding medium

b. The initial temperature of the object

c. The maximum temperature reached

d. The rate of heat transfer

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